Class Activity: 10.07.2026

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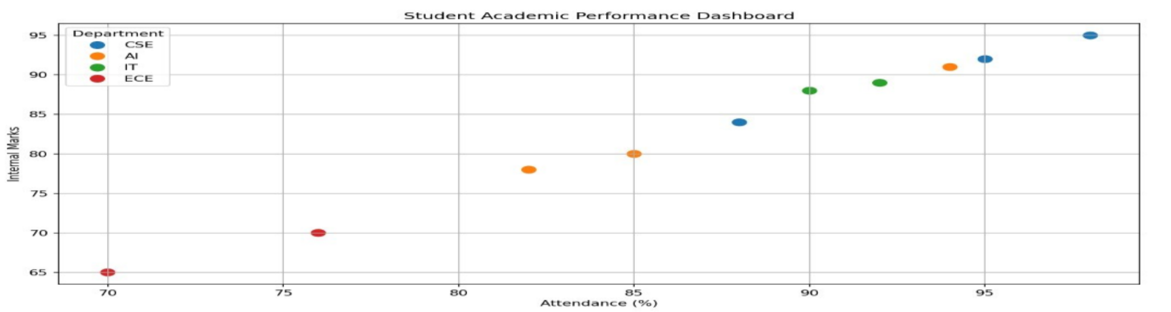
Scenario 1: Student Academic Performance Dashboard

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
data = {
   'Student_ID':['S101','S102','S103','S104','S105','S106','S107','S108','S109','S110'],
   'Attendance':[95,82,90,76,98,85,92,70,88,94],
   'Internal_Marks':[92,78,88,70,95,80,89,65,84,91],
   'Department':['CSE','AI','IT','ECE','CSE','AI','IT','ECE','CSE','AI'] }
df = pd.DataFrame(data)
plt.figure(figsize=(8,6))
sns.scatterplot(data=df,
                x='Attendance',
                y='Internal_Marks',
                hue='Department',
                s=120)
plt.title("Student Academic Performance Dashboard")
plt.xlabel("Attendance (%)")
plt.ylabel("Internal Marks")
plt.grid(True)
plt.show()
```

Output:



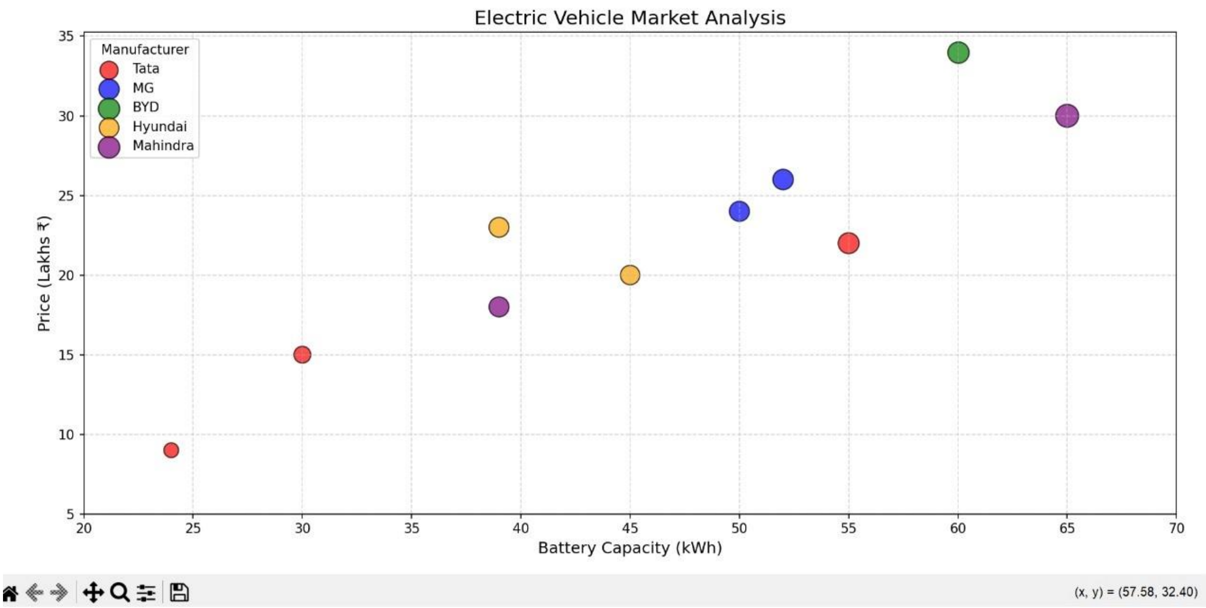
Scenario 2: Electric Vehicle Market Analysis

Code:

```
# Colors
colors = {
    'Tata':'red',
    'MG':'blue',
    'BYD':'green',
    'Hyundai':'orange',
    'Mahindra':'purple'
}
plt.figure(figsize=(9,6))
for company in colors:
    temp = df[df['Manufacturer'] == company]
    plt.scatter(
        temp['Battery_Capacity'],
        temp['Price'],
        s=temp['Driving_Range']/2,      # Smaller bubble size
        color=colors[company],
        edgecolors='black',             # Black border
        linewidth=1,
        alpha=0.7,
        label=company
    )
plt.title("Electric Vehicle Market Analysis", fontsize=15)
plt.xlabel("Battery Capacity (kWh)", fontsize=12)
```

```
plt.ylabel("Price (Lakhs ₹)", fontsize=12)

plt.xticks(range(20,71,5))
plt.yticks(range(5,36,5))
plt.grid(True, linestyle='--', alpha=0.5)
plt.legend(title="Manufacturer")
plt.tight_layout()
plt.show()
```

Output:



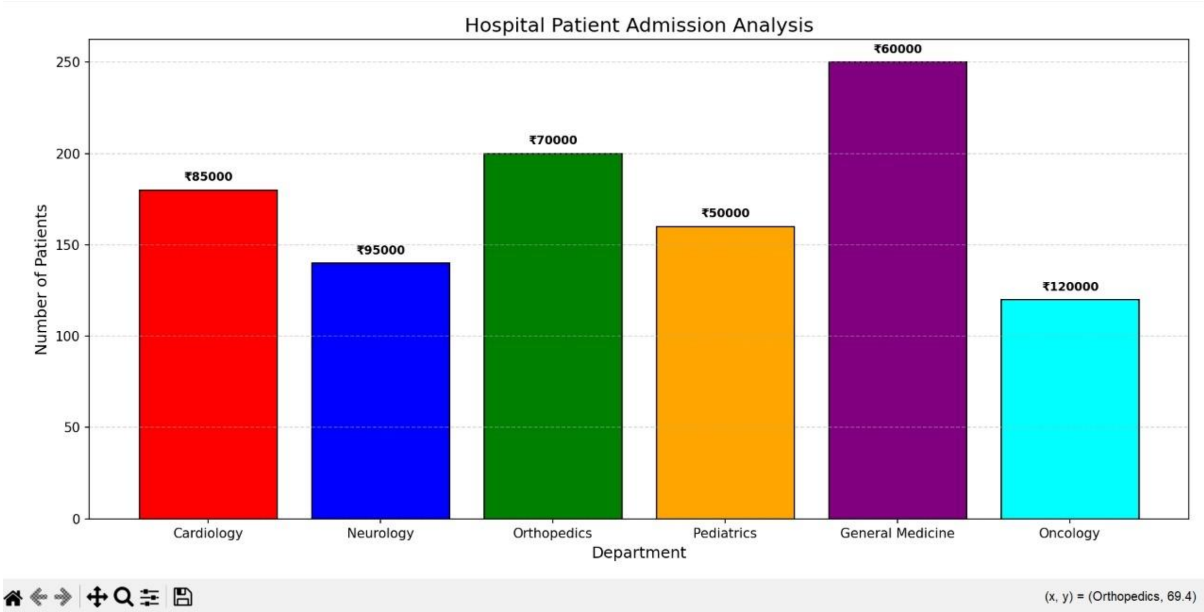
Scenario 3: Hospital Patient Admission Analysis

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
# Create Dataset
data = {
    'Department': ['Cardiology', 'Neurology', 'Orthopedics',
                   'Pediatrics', 'General Medicine', 'Oncology'],
    'Number_of_Patients': [180, 140, 200, 160, 250, 120],
    'Average_Treatment_Cost': [85000, 95000, 70000,
                               50000, 60000, 120000]
```

```
}
df = pd.DataFrame(data)
# Plot
plt.figure(figsize=(10,6))
colors = ['red', 'blue', 'green', 'orange', 'purple', 'cyan']

bars = plt.bar(df['Department'],
        df['Number_of_Patients'],
        color=colors,
        edgecolor='black')
# Display treatment cost on bars
for bar, cost in zip(bars, df['Average_Treatment_Cost']):
  plt.text(bar.get_x() + bar.get_width()/2,
       bar.get_height() + 5,
       f'₹{cost}',
       ha='center',
       fontsize=9,
       fontweight='bold')
plt.title("Hospital Patient Admission Analysis", fontsize=15)
plt.xlabel("Department", fontsize=12)
plt.ylabel("Number of Patients", fontsize=12)
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()
```

Output:



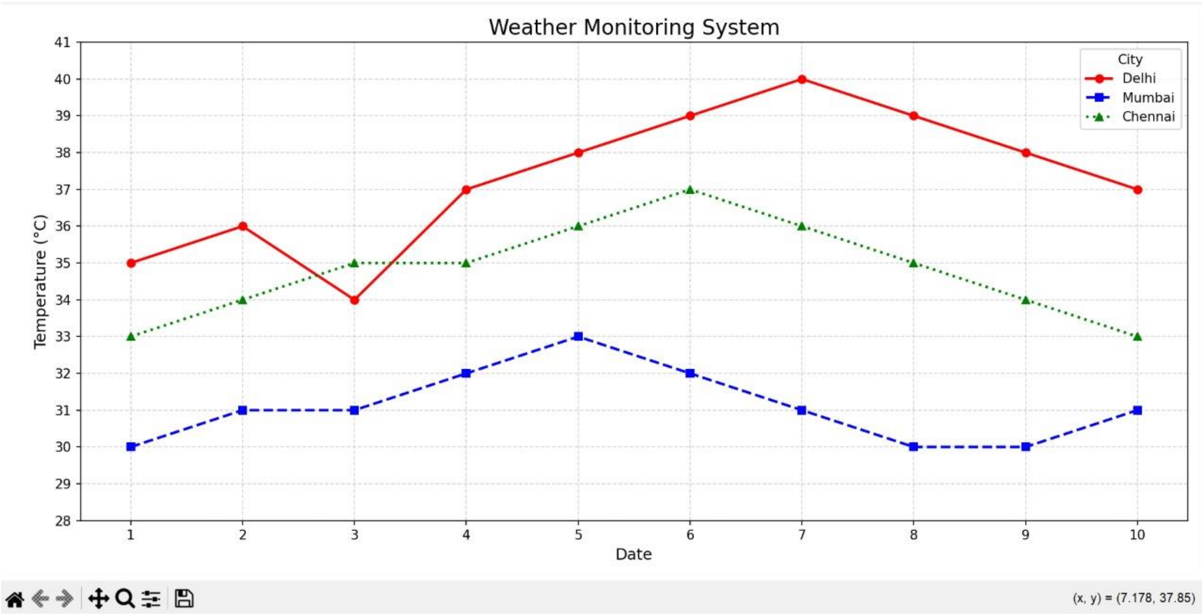
Scenario 4: Weather Monitoring System

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
# Create Dataset
data = {
    'Date': list(range(1, 11)),
    'Delhi': [35, 36, 34, 37, 38, 39, 40, 39, 38, 37],
    'Mumbai': [30, 31, 31, 32, 33, 32, 31, 30, 30, 31],
    'Chennai': [33, 34, 35, 35, 36, 37, 36, 35, 34, 33]
}
df = pd.DataFrame(data)
# Create Figure
plt.figure(figsize=(10,6))
# Delhi
plt.plot(df['Date'],
         df['Delhi'],
         color='red',
         linestyle='-',
```

```
        marker='o',
        linewidth=2,
        label='Delhi')
# Mumbai
plt.plot(df['Date'],
        df['Mumbai'],
        color='blue',
        linestyle='--',
        marker='s',
        linewidth=2,
        label='Mumbai')
# Chennai
plt.plot(df['Date'],
        df['Chennai'],
        color='green',
        linestyle=':',
        marker='^',
        linewidth=2,
        label='Chennai')
# Chart Details
plt.title("Weather Monitoring System", fontsize=16)
plt.xlabel("Date", fontsize=12)
plt.ylabel("Temperature (°C)", fontsize=12)
plt.xticks(range(1,11))
plt.yticks(range(28,42))
plt.grid(True, linestyle='--', alpha=0.5)
plt.legend(title="City")
plt.tight_layout()
plt.show()
```

Output:



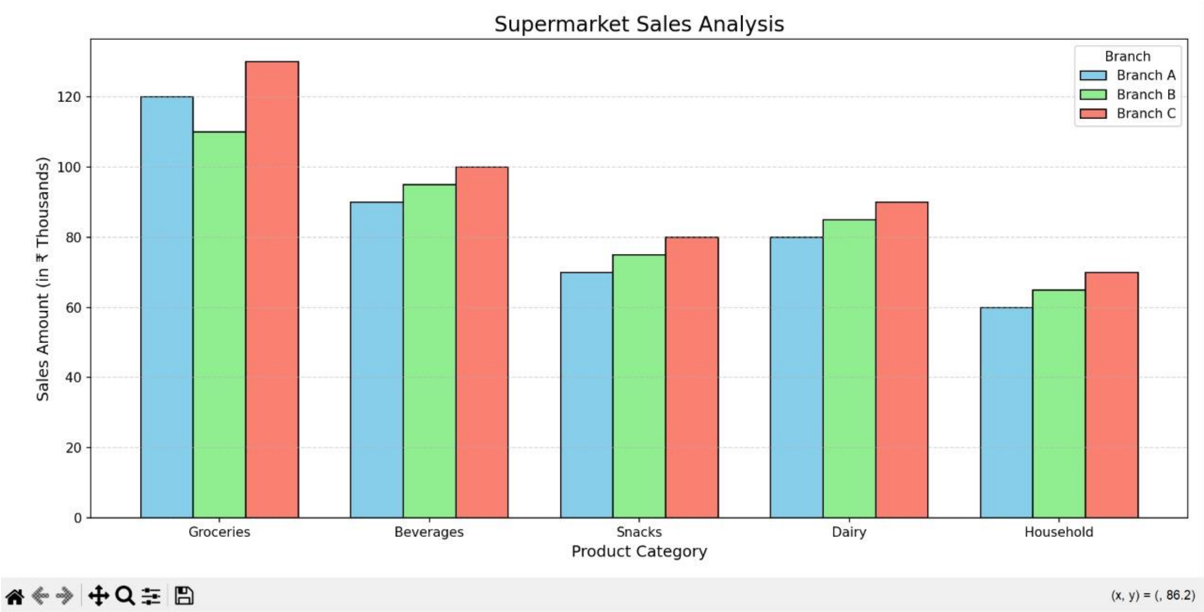
Scenario 5: Supermarket Sales Analysis

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

import numpy as np

# Create Dataset

data = {

    'Product_Category': ['Groceries', 'Beverages', 'Snacks', 'Dairy', 'Household'],

    'Branch_A': [120, 90, 70, 80, 60],

    'Branch_B': [110, 95, 75, 85, 65],

    'Branch_C': [130, 100, 80, 90, 70]

}

df = pd.DataFrame(data)

# Position of bars

x = np.arange(len(df['Product_Category']))

width = 0.25

plt.figure(figsize=(10,6))

# Grouped Bars

plt.bar(x - width,

        df['Branch_A'],
```

```
        width,
        label='Branch A',
        color='skyblue',
        edgecolor='black')
plt.bar(x,
        df['Branch_B'],
        width,
        label='Branch B',
        color='lightgreen',
        edgecolor='black')
plt.bar(x + width,
        df['Branch_C'],
        width,
        label='Branch C',
        color='salmon',
        edgecolor='black')
# Labels and Title
plt.title("Supermarket Sales Analysis", fontsize=16)
plt.xlabel("Product Category", fontsize=12)
plt.ylabel("Sales Amount (in ₹ Thousands)", fontsize=12)
plt.xticks(x, df['Product_Category'])
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.legend(title="Branch")
plt.tight_layout()
plt.show()
```

Output:



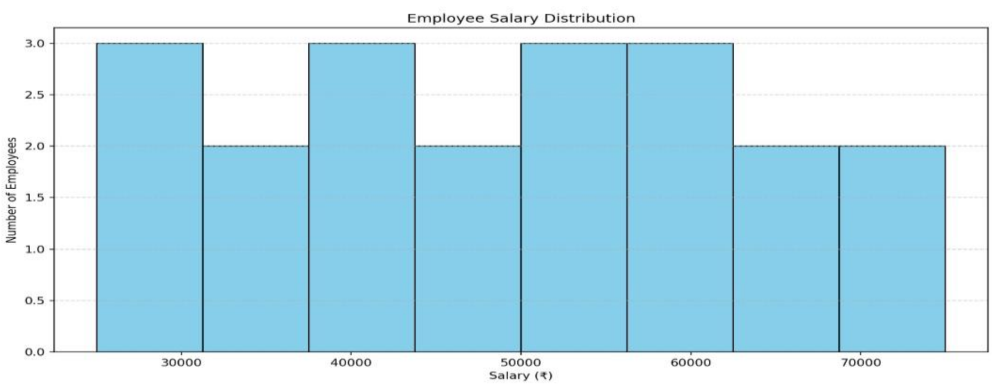
Scenario 6: Employee Salary Distribution

Code:

```
import matplotlib.pyplot as plt


# Employee Salary Dataset (in ₹)
salary = [
    25000, 28000, 30000, 32000, 35000,
    38000, 40000, 42000, 45000, 47000,
    50000, 52000, 55000, 58000, 60000,
    62000, 65000, 68000, 70000, 75000
]
# Create Histogram
plt.figure(figsize=(8,6))
plt.hist(
    salary,
    bins=8,
    color="skyblue",
    edgecolor="black"
)
# Title and Labels
```

```
plt.title("Employee Salary Distribution")
plt.xlabel("Salary (₹)")
plt.ylabel("Number of Employees")
# Grid
plt.grid(axis='y', linestyle='--', alpha=0.6)
plt.show()
```

Output:



Scenario 7: Smartphone Market Comparison

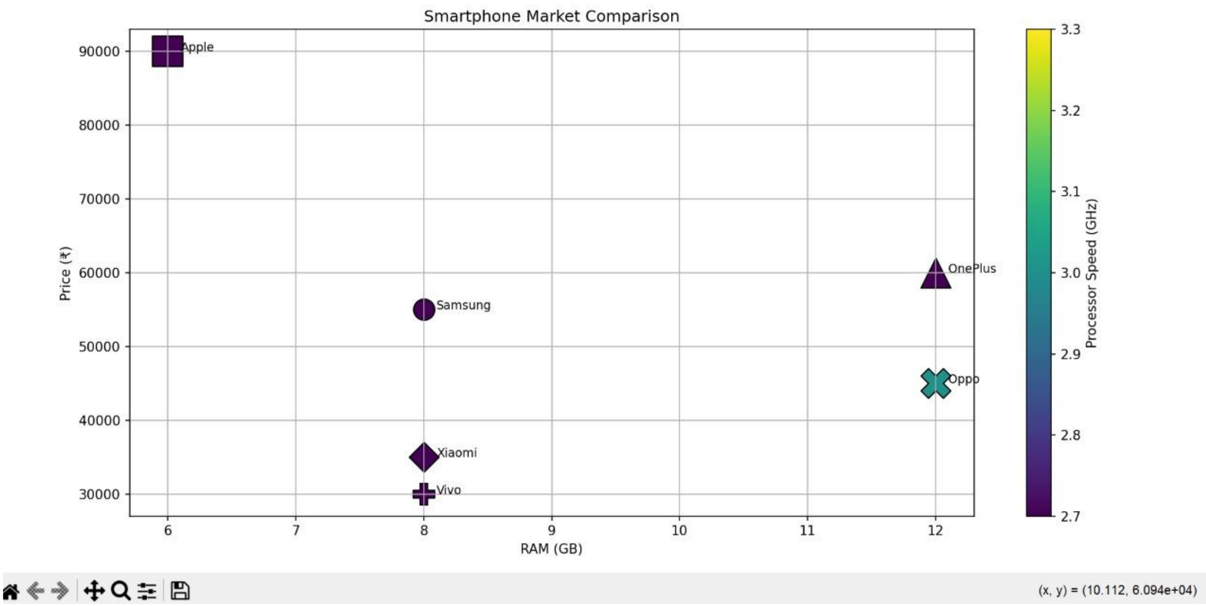
Code:

```
import pandas as pd
import matplotlib.pyplot as plt
# Dataset
data = {
    'Brand': ['Samsung','Apple','OnePlus','Xiaomi','Vivo','Oppo'],
    'RAM': [8,6,12,8,8,12],
    'Processor_Speed': [2.8,3.2,3.1,2.7,2.6,3.0],
    'Price': [55000,90000,60000,35000,30000,45000],
    'Storage': [128,256,256,128,128,256]
}
df = pd.DataFrame(data)
markers = {
    'Samsung':'o',
    'Apple':'s',
```

```
    'OnePlus':'^',
    'Xiaomi':'D',
    'Vivo':'P',
    'Oppo':'X'
}
plt.figure(figsize=(8,6))
# Plot each point
for i, row in df.iterrows():
    sc = plt.scatter(
        row['RAM'],
        row['Price'],
        s=row['Storage']*2,
        c=[row['Processor_Speed']],   # Notice the square brackets
        cmap='viridis',
        marker=markers[row['Brand']],
        edgecolors='black'
    )
    # Brand name beside each point
    plt.text(row['RAM']+0.1,
             row['Price'],
             row['Brand'],
             fontsize=9)
# Color Bar
plt.colorbar(sc, label="Processor Speed (GHz)")
# Labels
plt.title("Smartphone Market Comparison")
plt.xlabel("RAM (GB)")
plt.ylabel("Price (₹)")
plt.grid(True)
plt.tight_layout()
```

```
plt.show()
```

Output:



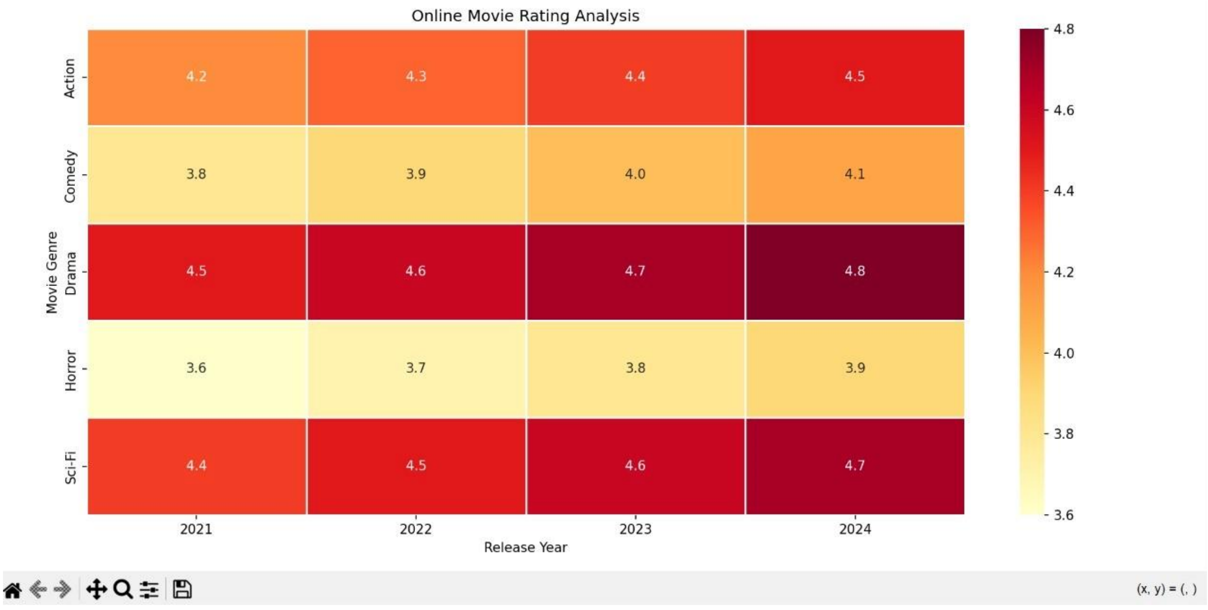
Scenario 8: Online Movie Rating Analysis

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Create Dataset
data = {
    'Genre': ['Action', 'Comedy', 'Drama', 'Horror', 'Sci-Fi'],
    '2021': [4.2, 3.8, 4.5, 3.6, 4.4],
    '2022': [4.3, 3.9, 4.6, 3.7, 4.5],
    '2023': [4.4, 4.0, 4.7, 3.8, 4.6],
    '2024': [4.5, 4.1, 4.8, 3.9, 4.7]
}
df = pd.DataFrame(data)
# Set Genre as index
df.set_index('Genre', inplace=True)
# Plot Heatmap
plt.figure(figsize=(8,6))
```

```
sns.heatmap(
    df,
    annot=True,
    cmap='YlOrRd',
    linewidths=0.5,
    fmt=".1f"
)
plt.title("Online Movie Rating Analysis")
plt.xlabel("Release Year")
plt.ylabel("Movie Genre")
plt.tight_layout()
plt.show()
```

Output:



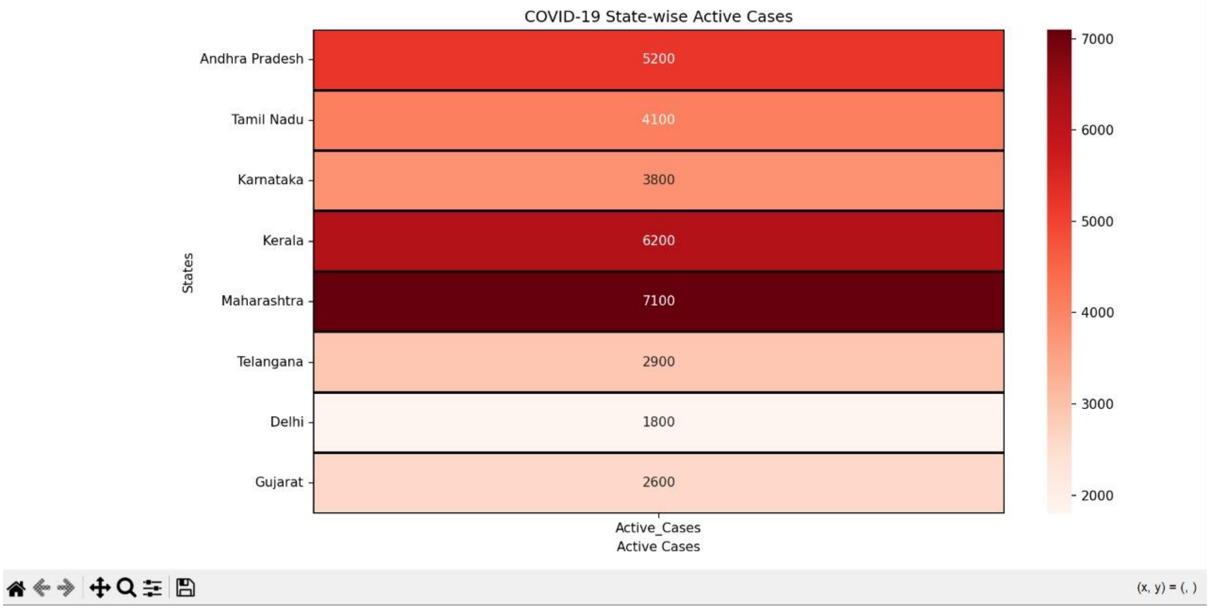
Scenario 9: COVID-19 State-wise Analysis

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
import seaborn as sns
# Create Dataset
data = {
    'State': ['Andhra Pradesh', 'Tamil Nadu', 'Karnataka',
              'Kerala', 'Maharashtra', 'Telangana',
              'Delhi', 'Gujarat'],

    'Active_Cases': [5200, 4100, 3800, 6200, 7100, 2900, 1800, 2600]
}
# Create DataFrame
df = pd.DataFrame(data)
# Set State as Index
df.set_index('State', inplace=True)

# Plot Heatmap (Tile Plot)
plt.figure(figsize=(6,6))
sns.heatmap(
    df,
    annot=True,
    cmap='Reds',
    linewidths=1,
    linecolor='black',
    fmt='d'
)

plt.title("COVID-19 State-wise Active Cases")
plt.xlabel("Active Cases")
plt.ylabel("States")
plt.tight_layout()
plt.show()
```

Output:



Scenario 10: Iris Flower Classification

Code:

```
from sklearn.datasets import load_iris

import pandas as pd

import matplotlib.pyplot as plt

# Load Built-in Iris Dataset

iris = load_iris()

# Create DataFrame

df = pd.DataFrame(

    iris.data,

    columns=['Sepal_Length', 'Sepal_Width',

             'Petal_Length', 'Petal_Width']

)

# Add Species Names

df['Species'] = [iris.target_names[i] for i in iris.target]

# Colors and Markers

colors = {

    'setosa': 'red',
```

```
    'versicolor': 'blue',
    'virginica': 'green'
}
markers = {
    'setosa': 'o',
    'versicolor': 's',
    'virginica': '^'
}
plt.figure(figsize=(8,6))
# Plot Each Species
for species in df['Species'].unique():
    temp = df[df['Species'] == species]
    plt.scatter(
        temp['Sepal_Length'],
        temp['Petal_Length'],
        s=temp['Sepal_Width'] * 40,      # Point Size
        color=colors[species],           # Color
        marker=markers[species],         # Shape
        edgecolors='black',
        alpha=0.7,
        label=species
    )
# Labels and Title
plt.title("Iris Flower Classification")
plt.xlabel("Sepal Length (cm)")
plt.ylabel("Petal Length (cm)")
# Legend
plt.legend(title="Species")
# Grid
plt.grid(True, linestyle='--', alpha=0.5)
```

plt.tight_layout()

plt.show()

Output: